MILITARY SPECIFICATION

ADJUSTER, LAP BELT, 1-3/4 INCH WEBBING

This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for lap belt adjusters. Adjusters are to be used with 1-3/4 inch webbing.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-4088	Webbing, Textile, Woven Nylon
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice) Process for
MIL-F-7179	Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts; General Specification for
MIL-W-19078	Webbing, Textile, Polyester, Latex Impregnated
MIL-L-23398	Lubricant, Solid Film, Air Drying
MIL-W-25361	Webbing, Textile, Polyester, Low Elongation

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing and quality conformance tests.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials and processes

3.3.1 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.1.1 Dissimilar metals. Dissimilar metals are defined in MIL-STD-889. If necessary, combinations of dissimilar metals having a relatively weak galvanic potential may be used provided their contact surfaces are separated and protected by a passive coating. Any dissimilar metal and protective coating combination proposed to be used shall be forwarded to the procuring activity for approval prior to fabrication or installation.

3.3.1.2 Protective treatment. When materials are subject to deterioration from climatic and environmental conditions likely to occur during service usage, they shall be protected in accordance with MIL-F-7179 against such deterioration in a manner that will in no way prevent compliance with the applicable performance requirements. The protective coating shall not crack, chip, or scale with age or extremes of climatic and environmental conditions.

3.3.1.3 Heat treatment. Steel components shall be heat treated in accordance with MIL-H-6875.

3.3.2 Lubrication. Lubrication of parts shall be accomplished in accordance with MIL-L-23398.

3.4 Design and construction. The adjuster, when installed on the lap belt, shall permit the crewman to adjust his lap belt properly without difficulty. (Two adjusters are required per lap belt.) The adjuster shall be designed to be compatible with lap belts made of 1-3/4-inch wide webbing in accordance with MIL-W-4088, MIL-W-19678, and MIL-W-25361 or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

3.4.1 Environmental conditions. The adjuster shall be designed to meet the operating requirements of this specification during exposure to the environmental conditions or natural combination of conditions specified in table I.

TABLE I. Environmental conditions..

Environmental Condition	Procedure & Method of MIL-STD-810
High temperature	I of 501.1
Low temperature	I of 502.1
Humidity	I of 507.1
Vibration	I of 514.2
Shock	III of 516.2
Temperature shock	I of 503.1
Fungus	I of 508.1
Dust (fine sand)	I of 510.1
Salt fog	I of 509.1

3.5 Performance

3.5.1 Slippage and creep. The adjuster shall provide the following features:

a. Ease of adjustment.

b. Shall not allow slippage more than one-half inch under 2,500 cycles of varying loads into webbing.

c. Shall not creep more than one-half inch while being worn, regardless of mission duration.

3.5.2 Proof load. The lap belt adjuster assembly shall withstand a proof load of 1,000 pounds, applied for a 30-second period, without deformation or failure of the components of the adjuster. During the application of the proof load, there shall be no slippage of webbing through the adjuster.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

3.5.3 Static load. The lap belt adjuster assembly shall withstand a static load of 2,880 pounds, applied for a 30-second period. During application of the static load, there shall be no slippage of webbing through the adjuster.

3.5.4 Dynamic load. Two adjusters, arranged in a lap belt configuration, shall be dynamically tested to 5,760 pounds for .01-second duration. During the application of the dynamic load, there shall be no slippage of the webbing through the adjuster.

3.5.5 Operation. With a 25-pound load applied at point B of the belt, a force not to exceed 50 pounds at scale C shall be required to adjust the belt (see figure 1). When this force is applied, the webbing shall move freely through the adjuster with 25 pounds at B in figure 1. If a tab is used, the adjuster release mechanism shall be operable with one hand pulling a lanyard parallel to the lap belt webbing and in a direction away from the lap belt anchor fitting (see figure 1 for direction). The force required to operate release tab shall not exceed 10 pounds.

3.6 Tolerances on dimensions. The adjuster shall not cause an interference fit (see 6.3.1).

3.7 Interchangeability. All parts having the same part number shall be interchangeable with each other with respect to installation and performance. Changes in part number shall be governed by the drawing number requirements of MIL-STD-100.

3.8 Identification of product. The adjuster shall be permanently identified by serial number, approved manufacturer's code symbol, and the last two digits of the year of manufacture in accordance with MIL-STD-130.

3.9 Workmanship. The adjuster shall be manufactured in a thoroughly workmanlike manner and shall be free of any defects that might affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the adjuster shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

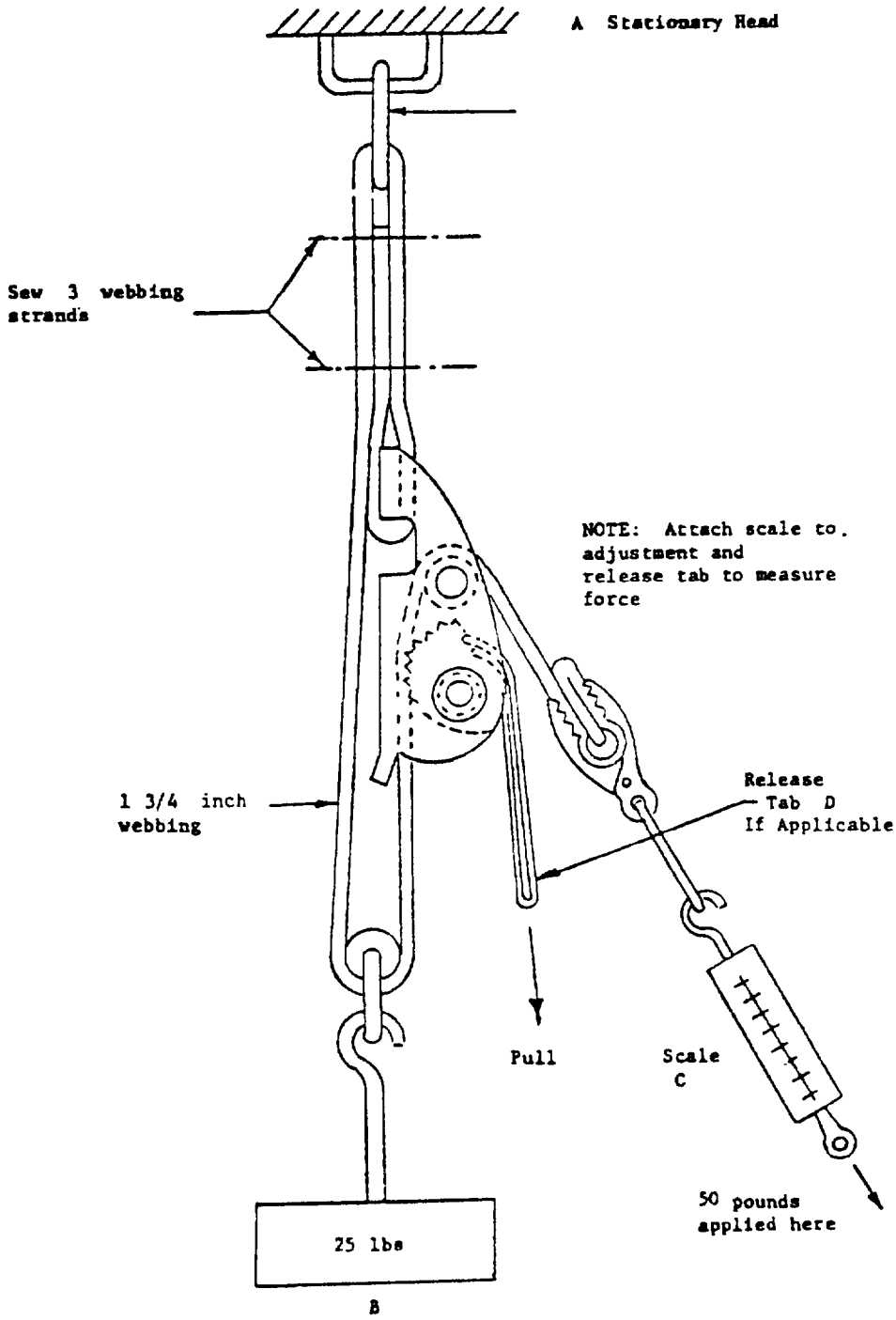


FIGURE 1. Test setup for operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

4.3 Preproduction tests. The preproduction samples shall be subjected to the preproduction tests which shall consist of all of the test methods described under 4.6, except that only two of the preproduction samples shall be subjected to the static load test specified in 4.6.5 and ten shall be used for the dynamic load test specified in 4.6.6.

4.3.1 Preproduction samples. The preproduction samples specified in table II shall consist of a minimum of 12 adjusters that are representative in every way of the production items. All preproduction samples that have successfully passed the preproduction tests shall be submitted to the procuring activity for engineering evaluation. The items shall be clearly identified as preproduction samples.

4.3.2 Preproduction test report. When specified by the procuring activity, a preproduction test report shall be submitted to the procuring activity. The order of testing shall be included in the preproduction test report.

4.3.3 Rejection and retest (preproduction). If the preproduction sample fails to pass any preproduction test requirement, that preproduction test shall be stopped and the necessary corrective action taken. Full particulars of the failure and the corrective action taken as well as subsequent test results shall be reported to the procuring activity for approval.

4.4 Quality conformance tests. Quality conformance tests shall be performed on production items.

4.4.1 Individual tests. Each adjuster shall be tested by the contractor. As a minimum, the following tests, as described under 4.6, shall be conducted:

a. Examination of product (see 4.6.1)

b. Proof load (see 4.6.2)

c. Magnaflux (see 4.6.3)

d. Operation (see 4.6.8).

4.4.2 Tests by the procuring activity. The procuring activity shall have the option to conduct tests on production items on a random basis. These tests will not exceed the requirements of this specification.

4.4.3 Rejection and retest (production items). Failure of any production item to pass any of the quality conformance tests shall be cause for rejection of the item. If a production item is resubmitted, full particulars concerning the failure and corrective action taken shall be submitted with the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

TABLE II. Preproduction test schedule.

Test	Test Paragraph Number	Unit Number 1	2	3	4	5	6	7	8	9	10	11	12
Examination of product	4.6.1	x	x	x	x	x	x	x	x	x	x	x	x
Proof load	4.6.2	x	x	x	x	x	x	x	x	x	x	x	x
Magnaflux	4.6.3	x	x	x	x	x	x	x	x	x	x	x	x
Slippage and creep	4.6.4					x		x	x		x	x	x
Lap belt tension	4.6.8.1			x				x		x		x	
Lap belt release	4.6.8.2		x		x		x		x		x		
Static load	4.6.5											x	x
Dynamic load	4.6.6	x	x	x	x	x	x	x	x	x	x		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

4.5 Test condition

4.5.1 Atmospheric conditions. Unless otherwise specified all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 25° ±10°C (77° ±18°F) and at a relative humidity of 80 percent or less.

4.5.2 Adjuster. Unless otherwise specified, all tests shall be conducted with the webbing reeved through the adjuster as shown in figure 2.

4.5.3 Test unit. Unless otherwise specified, inspection and testing shall be conducted on complete assemblies.

4.6 Test methods

4.6.1 Examination of product. The adjuster shall be subjected to a visual examination to determine compliance with the requirements of this specification:

a. The adjuster shall be examined for compliance with the requirements of identification marking specified in this specification and MIL-STD-130.

b. The adjuster shall be measured with suitable certified measuring instruments to determine whether it conforms to the dimensions of the applicable drawings and of this specification and to insure interchangeability and compatibility of mating components.

c. The examination shall assure that the adjuster is free of pits, burrs, fins, sharp edges, and excess porosity.

4.6.2 Proof loads. The lap belt adjuster assembly shall be subjected to a 1,000 pound proof load, with the load applied through points A and B as indicated on figure 2, and with a minimum of 2 inches of free webbing between the adjuster and the adjustment strap end. The full proof load shall be maintained for a minimum of 30 seconds. There shall be no breakage or deformation of the hardware nor more than one-half inch slippage of webbing through the adjuster. Any adjuster rejected as a result of this test shall not be submitted for retest.

4.6.3 Magnaflux. Prior to plating and assembly and after heat treatment, all load-carrying members of the adjuster shall be checked 100 percent by magnafluxing on a suitable magnaflux machine or other suitable equipment in conformance with MIL-H-6868. Evidence of any cracks in any areas, which could affect the strength of the adjuster, shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

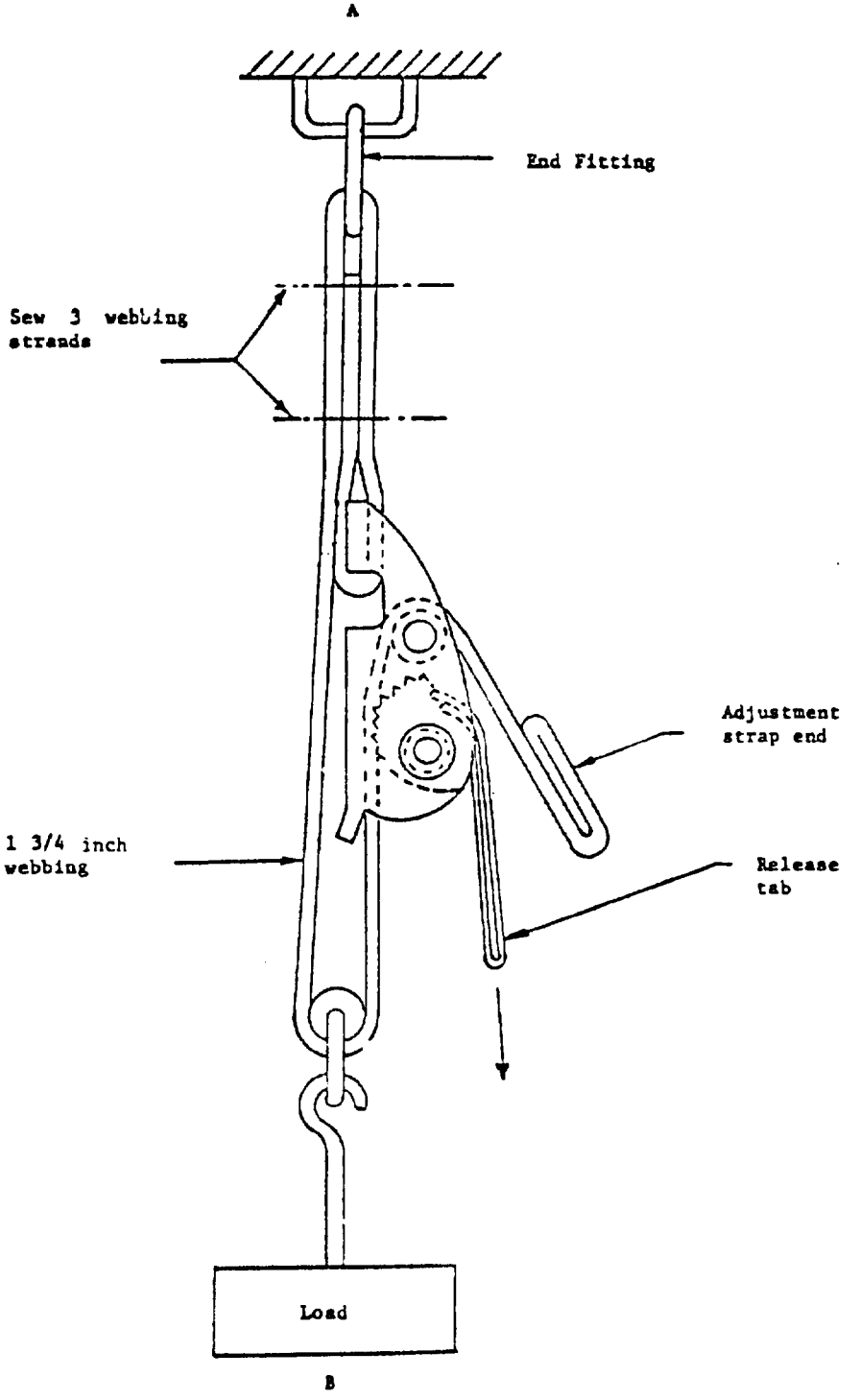


FIGURE 2. Test setup for release tab.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

4.6.4 Slippage and creep. A cycle consisting of varying loads of 160 pounds and zero pounds shall be applied for 2,500 cycles. The test shall be started by pulling the webbing through the adjuster to eliminate any set in the webbing or false load of the test equipment. The 20-pound load and the zero pound load shall be at a stroke of at least three-fourths of an inch. At the end of the 2,500 cycles, the webbing shall not have slipped or crept more than one-half inch. The rate of loading shall not be greater than 60 cycles per minute (see figure 2 for application of load).

4.6.5 Static load. The lap belt adjuster assembly shall be subjected to a 2,880 pound static load (in tension) for 30 seconds to determine structural integrity. The load syall be applied at a rate of 1 inch per minute travel through points A and B as shown on figure 2 and with a minimum of 2 inches of free webbing between the adjuster and the adjustment strap end. Deformation is permitted. During this test there shall be no more than one-half inch of webbing slippage through the adjuster. Any adjuster rejected as a result of this test shall be submitted for retest. This is a destructive test; therefore, any adjuster subjected to this test shall be tagged to indicate it is not to be used for service.

4.6.6 Dynamic load. Ten adjusters shall be assembled, two per belt, and subjected to a dynamic load of 5,760 pounds for .01 second duration. The test shall be conducted on a horizontal or vertical decelerator with a simulated seat installation. This is a destructive test; therefore, any adjuster subject to this test shall be tagged to indicate it is not to be used for service.

4.6.7 Exposure to environmental conditions. The adjuster shall be subjected to the environmental tests specified in table I. After each of the environmental tests, the adjuster shall pass the tests specified in 4.6.8.1 and 4.6.8.2.

4.6.8 Operation

4.6.8.1 Lap belt tension. The adjuster shall be subjected to the following lap belt tension test using the configuration shown in figure 1: With point A anchored and point B attached to a 25-pound weight, apply a load to the lap belt adjuster. The force required to pull the webbing through the adjuster shall not exceed 50 pounds at point C.

4.6.8.2 Lap belt release. If a tab is used, the adjuster shall be subjected to the following lap belt release test: With point A anchored, and a load at B (see figure 1), apply a force to the adjuster release tab D. While this force is applied, the webbing shall move freely through the adjuster. The force required to operate the release tab shall not exceed 10 pounds.

5. PACKAGING

5.1 Packaging shall be in accordance with the instructions of the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The adjuster covered by this specification is intended to be used to adjust any of the lap belts made of the 1 3/4 inch wide webbing.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Preproduction requirements (see 4.3.2).

c. Packaging.

6.3 Definitions

6.3.1 Interference fit. Lap belt drawings normally show webbing constructed in accordance with a specified width and reeved through slots in the hardware. Deformation of this webbing because of insufficient width of slots or excessive internal slot radii shall be considered an interference fit.

6.3.2 Pits. Cavities in a metal surface of a component caused by non-uniform plating, by corrosion, by gas pitting, or by oxidation of forging material residue in the forging die shall be considered pits.

6.3.3 Porosity. Numerous small holes distributed through the metal, in either casting or plating material, shall be considered porosity.

6.3.4 Cracks. In a casting, separations which are caused by residual stress concentration during cooling, due to the shape of the objects shall be considered cracks.

6.3.5 Sharp edges, burrs, and fins. Sharp edges, burrs, and fins shall be considered defects resulting from machining, filing, grinding, rolling, forging, or stamping of metal.

Custodian:
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Air Force - 11

Project No. 1680-F427

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:14Z
Check the source to verify that this is the current version before use.